

3.4 (Continue)

$s(t)$: position as function of t moving object

Velocity $v(t) = s'(t) = \frac{ds}{dt}$

Acceleration $a(t) = v'(t) = s''(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$

Jerk $j(t) = a'(t) = v''(t) = s'''(t)$

$$\frac{d^2s}{dt^2} = \frac{d}{dt} \left(\frac{ds}{dt} \right) = \frac{d^2s}{(dt)^2} = \frac{d^2s}{dt^2} \quad (\text{m.s}^{-2})$$

Economics

Derivative = margin

Let X = #units of product produced

$C(X)$ = cost of producing X units of the product

Def Marginal cost of production
 $= C'(X) = \frac{dC}{dX} = \lim_{h \rightarrow 0} \frac{C(X+h) - C(X)}{h}$

If h is small, $C'(X) \approx \frac{C(X+h) - C(X)}{h}$

$$C(X+h) - C(X) \approx C'(X) \cdot h$$

If X is huge, $h=1$ is considered to be small

$$C(X+1) - C(X) \approx C'(X)$$

Cost of producing 1 more unit of the product when X units of product has been produced

Ex. 5 $C(X) = X^3 - 6X^2 + 15X$

$$C'(X) = 3X^2 - 12X + 15$$

$$C'(10) = 300 - 120 + 15 = 195 = \text{cost of producing 1 more radiator when 10 units have already been produced}$$

$$C(11) - C(10) = (11^3 - 6(11)^2 + 15(11)) - (10^3 - 6(10)^2 + 15(10))$$

$$= 770 - 550 = 220$$

3.5 Derivative of Trigonometry function

$$(\sin X)' = \lim_{h \rightarrow 0} \frac{\sin(X+h) - \sin X}{h} = \lim_{h \rightarrow 0} \frac{\sin X \cos h + \sin h \cos X - \sin X}{h} = \lim_{h \rightarrow 0} \frac{\sin h \cos X}{h} + \lim_{h \rightarrow 0} \frac{\sin X (\cos h - 1)}{h}$$

$$= 1 \cdot \cos X + 0 = \cos X$$

$$(\cos X)' = \lim_{h \rightarrow 0} \frac{\cos(X+h) - \cos X}{h} = \lim_{h \rightarrow 0} \frac{\cos X \cos h - \sin X \sin h - \cos X}{h} = \lim_{h \rightarrow 0} \frac{\cos X (\cos h - 1)}{h} - \lim_{h \rightarrow 0} \frac{\sin X \sin h}{h}$$

$$= 0 - \sin X = -\sin X$$

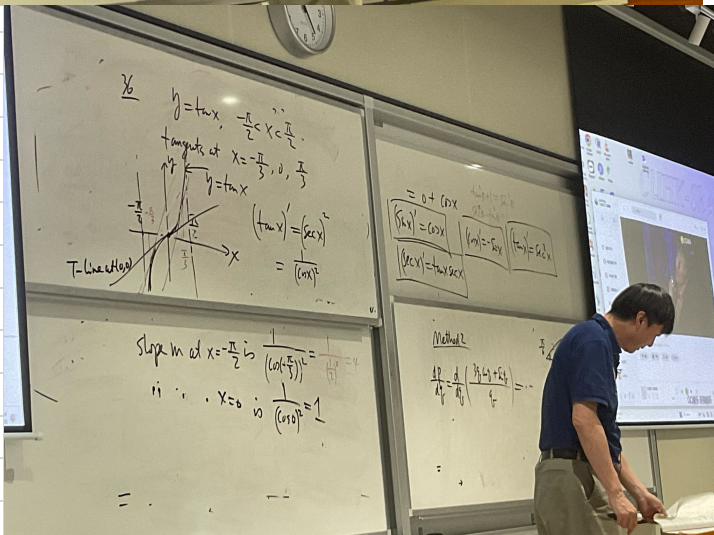
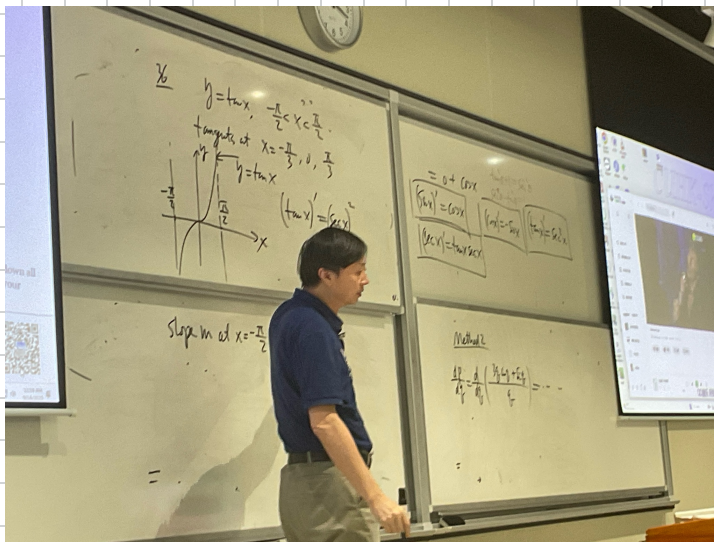
Fourier's series $\rightarrow$ use trig func to estimate other functions

$$(\tan X)' = \left(\frac{\sin X}{\cos X} \right)' = \frac{\cos X \cdot \cos X - \sin X \cdot (-\sin X)}{(\cos X)^2} = \frac{1}{\cos^2 X} = \sec^2 X$$

$$(\csc X)' = \left(\frac{1}{\sin X} \right)' = \frac{\sin X \cdot 0 - 1 \cdot \cos X}{\sin^2 X} = -\csc X \cot X$$

$$(\sec X)' = \left(\frac{1}{\cos X} \right)' = \frac{\cos X \cdot 0 - 1 \cdot (-\sin X)}{\cos^2 X} = \sec X \tan X$$

$$32. \quad p = \frac{3q + \tan q}{q \sec q}, \quad \frac{dp}{dq} = \frac{q \sec q \cdot (3 \sec^2 q) - (3q + \tan q)(q \sec q \tan q + \sec q)}{(q \sec q)^2}$$



3.6 Chain rule

$$y = (x^2 + 1)^2$$

$$\frac{dy}{dx} = \frac{d}{dx} (x^4 + 2x^2 + 1) = 4x^3 + 4x$$

$$y = (x^2 + 1)^{821} \rightarrow \frac{dy}{dx} = ?$$

$$\text{Let } g(x) = x^2 + 1 \rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\begin{aligned} y &= f(u) = u^{821} \\ u &= g(x) = x^2 + 1 \\ \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= 821 u^{820} \cdot 2x \\ &= 821 (x^2 + 1)^{820} \cdot 2x \end{aligned}$$

Use chain rule

$$\text{Consider } y = f \circ g(x) = f(g(x))$$

Suppose $g(x)$ is differentiable at c

Then $f \circ g(x)$ is differentiable at $x=c$ and
 $(f \circ g)'(c) = f'(g(c)) \cdot g'(c)$

In Leibnitz's notation: Write $u = g(x)$, $y = f(u)$

$$\text{Chain: } x \xrightarrow{g} u \xrightarrow{f} y \quad y = f(u) = f(g(x))$$

$$\left. \frac{dy}{dx} \right|_{x=c} = \left. \frac{dy}{du} \right|_{u=g(c)} \cdot \left. \frac{du}{dx} \right|_{x=c}$$

$$\text{Proof: } \left. \frac{dy}{dx} \right|_{x=c} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$$

$$\Delta y = f(g(c+\Delta x)) - f(g(c))$$

$$\Delta u = g(c+\Delta x) - g(c)$$

$$= \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(g(c+\Delta x)) - f(g(c))}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{g(c+\Delta x) - g(c)}{\Delta x}$$

By continuity of g at c ,
 $\Delta u \rightarrow 0$ as $\Delta x \rightarrow 0$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(g(c)+\Delta u) - f(g(c))}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{g(c+\Delta x) - g(c)}{\Delta x}$$

$$= \left. \frac{dy}{du} \right|_{u=g(c)} \cdot \left. \frac{du}{dx} \right|_{x=c}$$

3.6/80

$$y = \sqrt{x^2 + 7}$$

$$y = \sqrt{u} \quad u = x^2 + 7$$

T-Line at $x=2$

$$m = \left. \frac{dy}{dx} \right|_{x=2} = \left. \frac{dy}{du} \right|_{u=g(2)} \cdot \left. \frac{du}{dx} \right|_{x=2}$$

$$= \frac{1}{2} \cdot u^{-1/2} \Big|_{u=9} \cdot (2x-1) \Big|_{x=2}$$

$$= \frac{1}{2} \cdot 9^{-1/2} \cdot 3$$

$$= \frac{1}{2} \cdot \frac{1}{3} \cdot 3 = \frac{1}{2}$$

$$(y-3) = \frac{1}{2}(x-2)$$

$$y = \frac{1}{2}x + 2$$

3.6/50

$$y = \cos\left(5\sin\left(\frac{x}{3}\right)\right)$$

$$y' = -\sin\left(5\sin\left(\frac{x}{3}\right)\right) \cdot 5\cos\left(\frac{x}{3}\right) \cdot \frac{1}{3}$$

$$= -\frac{5}{3}\sin\left(5\sin\left(\frac{x}{3}\right)\right)\cos\left(\frac{x}{3}\right)$$

3.6/74

x	f(x)	g(x)	f'(x)	g'(x)
0	1	1	5	$\frac{1}{3}$
1	3	-4	$-\frac{1}{3}$	$-\frac{8}{3}$

e) $g(f(x))'|_{x=0}$

$$= g'(f(0)) \cdot f'(0)$$

$$= g'(1) \cdot 5$$

$$= -\frac{8}{3} \cdot 5 = -\frac{40}{3}$$

f) $[(x'' + f(x))^{-2}]'|_{x=1}$

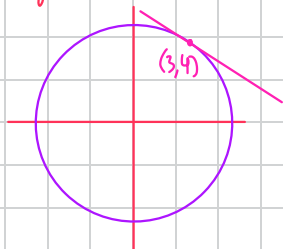
$$= -2(x'' + f(x))^{-3} \cdot (11x'' - f'(x))$$

$$= -2(1+3)^{-3} \cdot (11 - \frac{1}{3})$$

$$= -2 \cdot \frac{1}{64} \cdot \frac{32}{3} = -\frac{1}{3}$$

3.7 Implicit differentiation

e.g. $x^2 + y^2 = 5^2$ Want: equation of tangent line passing through (3,4)



$$y = \pm \sqrt{25 - x^2}$$

$$m = \frac{dy}{dx} \sqrt{25 - x^2} \Big|_{x=3}$$

$$= \frac{1}{2\sqrt{25-x^2}} (-2x) \Big|_{x=3}$$

$$= \frac{1}{2 \cdot 4} (-6) = -\frac{3}{4}$$

$$y - 4 = -\frac{3}{4}(x - 3)$$

$$y = -\frac{3}{4}x + \frac{25}{4}$$

Line equation with
slope m and passes
(x_0, y_0):

$$\frac{y - y_0}{x - x_0} = m$$

$$y = y_0 + m(x - x_0)$$

Smarter method

In $x^2 + y^2 = 25$

⊙, think y as a function of x

$$\frac{d}{dx} \odot = \frac{dx^2}{dx} + \frac{dy^2}{dx} = \frac{d25}{dx}$$

$$2x + \frac{dy^2}{dx} \cdot \frac{dy}{dx} = 0$$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$2y \cdot \frac{dy}{dx} = -2x \Rightarrow \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

$$m = \frac{dy}{dx} \Big|_{x=3} = -\frac{3}{4}$$

Prob. 36 $X \sin 2Y = Y \cos 2X$ At $(\frac{\pi}{4}, \frac{\pi}{2})$ Cannot solve for Y explicitly.

Goal: eq. of T-line

Check to see if the point is on the curve

$$\frac{\pi}{4} \cdot \sin \pi = \frac{\pi}{2} \cos \frac{\pi}{2}$$

$$0 = 0$$

$$\frac{d}{dx}(X \sin 2Y) = \frac{d}{dx}(Y \cos 2X)$$

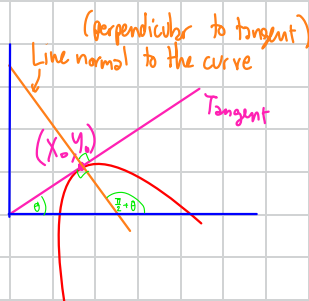
$$1 \cdot \sin 2Y + X \frac{d}{dx} \sin 2Y = \frac{dY}{dx} \cos 2X + Y \cdot \frac{d}{dx} \cos 2X$$

$$\sin 2Y + X(\cos 2Y) \cdot 2 \cdot \frac{dY}{dx} = \frac{dY}{dx} \cos 2X - Y \cdot \sin 2X \cdot 2$$

$$\frac{dY}{dx}(2X \cos 2Y - \cos 2X) = -2Y \sin 2X - \sin 2Y$$

$$\frac{dY}{dx} = \frac{-2Y \sin 2X - \sin 2Y}{2X \cos 2Y - \cos 2X}$$

$$\frac{dY}{dx} \text{ at } (\frac{\pi}{4}, \frac{\pi}{2}) = \frac{-2(\frac{\pi}{2})(1) - 0}{2(\frac{\pi}{4})(-1) - 0} = 2$$



$$\begin{aligned} \text{Slope} &= \tan(\theta + \frac{\pi}{2}) = \frac{\sin(\theta + \frac{\pi}{2})}{\cos(\theta + \frac{\pi}{2})} \\ &= \frac{\sin \theta \cos \frac{\pi}{2} + \cos \theta \sin \frac{\pi}{2}}{\cos \theta \cos \frac{\pi}{2} - \sin \theta \sin \frac{\pi}{2}} = -\frac{\cos \theta}{\sin \theta} = -\frac{1}{\tan \theta} \end{aligned}$$

$$\text{Slope of normal line} = -\frac{1}{m}$$

$$\text{Eq. of N-line: slope} = -\frac{1}{m} = -\frac{1}{2}$$

$$Y = \frac{\pi}{2} + \frac{1}{2}(X - \frac{\pi}{4}) = -\frac{1}{2}X + \frac{5\pi}{8}$$

Remark 1 If $m = 0$, so T-line is horizontal $\rightarrow$ N-line is vertical

(N-line's equation is $X = X_0$)

Remark 2 If T-line is vertical ($m = \pm \infty$) $\rightarrow$ N-line is horizontal ($Y = Y_0$)

3.8 Related rates

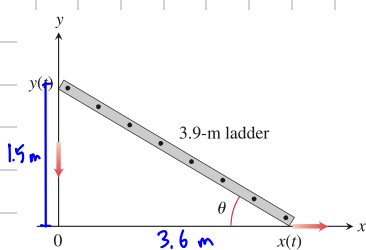
Scenario: Applied problems (physical, chemical, geometrical, biological, financial, etc.) involving independent variable t , dependent variable X, Y, Z, U, V

These variables are related by an equation such as $X^{423} + Y^{821} = 2025t$

Suppose we are given

$$\frac{dY}{dt}, \frac{dZ}{dt}, \frac{dU}{dt}, \frac{dV}{dt} \rightarrow \text{want } \frac{dX}{dt}$$

23.



$$x^2 + y^2 = 3.9^2$$

Know: $\frac{dx}{dt} = 1.5 \text{ m/s}$ when $x = 3.6 \text{ m}$
 Want: $\frac{dy}{dt}$, $\frac{d}{dt}(\frac{1}{2}xy)$, $\frac{d\theta}{dt}$

$$a) \frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}(3.9^2)$$

$$\frac{dx}{dt} \cdot 2x + \frac{dy}{dt} \cdot 2y = 0$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

At

$$\frac{dy}{dt} = \frac{-x \frac{dx}{dt}}{y} \Big|_{x=3.6} = \frac{-3.6 \cdot 1.5}{\sqrt{3.9^2 - 3.6^2}} = -3.6 \text{ m/s} \rightarrow 3.6 \text{ m/s down the wall}$$

$$b) \frac{dA}{dt} = \frac{1}{2} \frac{d}{dt}(xy) = \frac{1}{2} \left(x \frac{dy}{dt} + y \frac{dx}{dt} \right) \Big|_{x=3.6} = \frac{1}{2} (3.6 \cdot (-3.6) + 1.5 \cdot (1.5)) = \frac{1}{2} (-10.71) = -5.355 \text{ m}^2/\text{s}$$

$$c) \text{ From } \cos \theta = \frac{x}{3.9} \quad \frac{d}{dt}(\cos \theta) = \frac{d}{dt} \left(\frac{x}{3.9} \right) \\ -\sin \theta \frac{d\theta}{dt} = \frac{1}{3.9} \frac{dx}{dt} \\ \frac{d\theta}{dt} = -\frac{1}{3.9 \sin \theta} \cdot \frac{dx}{dt} = -\frac{1}{1.5} \cdot 1.5 = -1 \text{ rad/s}$$

Recipe

Step 1: Draw a diagram & Label all the variables

Step 2: Write down what you know and what you want

Step 3: Find equation relating variables

Step 4: $\frac{d}{dt}$ with respect to independent variable t

Step 5: Use what you know to find what you want

30. Know: $\frac{dV}{dt} \propto S$

$$V: \text{Volume} = \frac{4}{3}\pi r^3$$

$$S: \text{Surface area} = 4\pi r^2$$

r: radius

$$\therefore \frac{dV}{dt} = cS$$

Want to prove: $\frac{dr}{dt} = \text{const}$

$$\frac{dV}{dt} = \frac{d}{dt} \left(\frac{4}{3}\pi r^3 \right) = \frac{4}{3}\pi (3r^2) \frac{dr}{dt} = S \frac{dr}{dt}$$

$$\frac{dV}{dt} = cS$$

$$cS = S \frac{dr}{dt}$$

Q.E.D. (Quite easily done)

HW 3.5-3.8 I₂

3.9 Linearization of differentials

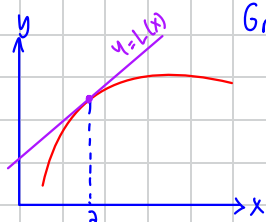
Recall: if f differentiable at a then $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$

$$\therefore \frac{f(x) - f(a)}{x - a} \approx f'(a) \text{ if } x \approx a \text{ (} x \neq a \text{)}$$

$$f(x) - f(a) \approx f'(a)(x - a) \text{ if } x \approx a$$

$$f(x) \approx \underbrace{f(a) + f'(a)(x - a)}_{L(a) \rightarrow \text{linear function}} \rightarrow \text{can be used in research}$$

Graph of $y = L(x)$ is T-line passing through $(a, f(a))$



Ex 1 $f(x) = \sin x, a = 0$
 $L(x) = \sin 0 + \cos 0 (x - 0)$
 $= 0 + 1(x) = x$

$$\rightarrow \sin x \approx 0 \text{ if } x \approx 0 \quad \left(\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \rightarrow \frac{\sin \theta}{\theta} \approx 1 \rightarrow \sin \theta \approx \theta; \theta \approx 0 \right)$$

Ex 2 $f(x) = \tan x, a = 0$
 $L(x) = \tan 0 + \sec^2(0)(x - 0) = x \rightarrow \tan x \approx x; x \approx 0$

$$\lim_{x \rightarrow 0} \frac{\tan(821x)}{\sin(423x)} \stackrel{\text{Secretly}}{\approx} \frac{821x}{423x}$$

$$= \frac{\sin(821x)}{\cos(821x) \sin(423x)} = \lim_{x \rightarrow 0} \frac{\frac{\sin(821x)}{821x}}{\cos(821x) \frac{\sin(423x)}{423x}} \cdot \frac{821x}{423x} = \frac{1}{1 \cdot 1} \cdot \frac{821x}{423x} = \frac{821}{423}$$

$\cos 0 = 1$